Name: _____

Date: _____

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Your Card Toolkit — Levels 1 to 10

Everything you already know, and what it does for your card

You have finished ten levels of Game Lab. That is enough to build a card that looks great and changes every time it runs. Movement and clicks join it later, after Levels 12 to 17.

#	Level	What you learned	The code	On your card
1	Programming for a Purpose	Good designs start with **who they are for**.	—	Your Criterion A brief names one person. Every choice serves them.
2	Plotting Shapes	The **400 × 400 grid**. `x` counts across from the left, `y` counts **down** from the top.	centre is `(200, 200)`	Sketch your card on the grid first, then read the numbers off your sketch.

3	Drawing in Game Lab	Drawing shapes and choosing their colour.	`fill("gold");`
			`rect(x, y);`
			`ellipse(x, y);`
				Each `fill` colours every shape after it, until the next `fill`.

4	Shapes and Parameters	**Parameters** set the width and height. A background fills the whole canvas.	`background("lightblue");`
			`rect(x, y, w, h);`
			`ellipse(x, y, w, h);`
				Background first, then the scenery: a cake from rectangles, balloons from ellipses, a sea from one wide rectangle.

5	Variables	A **variable** labels a number so you can use it again.	`var cakeX = 200;`	Build an object around one variable. Change that one number and the whole object moves.
6	Random Numbers	Programs that are **different every time** they run.	`randomNumber(0, 400)`	Confetti, stars or snow in new places on every Run, so the card feels alive.
7	Mini-Project: Robot Faces	Shapes, variables and randomness combined in one design: plan, try, then reflect.	—	The same process for your card: plan on paper, build a part, run it, improve it.

8	Sprites	A **sprite** is a picture you can control. Its image comes from the **Animation tab**.	`var boat =`
			`createSprite(x, y);`
			`boat.setAnimation("boat");`
			`drawSprites();`
				The characters and objects on your card: drawn by you, or picked from the library.

9	Sprite Properties	**Dot notation** changes a sprite's size, angle and position.	`boat.scale = 0.5;`
			`boat.rotation = 15;`
			`boat.x = 120;`
				Resize and tilt every picture so the card looks composed, not cluttered.

10	Text	Words on the canvas, placed and sized like a shape.	`textSize(32);` `text("Hello", x, y);`	Your message, big and clear. Its colour comes from the `fill` just before it.

Start your project

1. code.org → **Projects** → **Game Lab**. Rename the project **⟨Your Name⟩ Digital Greeting Card**.
2. Delete the three lines `function draw() { }`. They are the draw loop from Level 12; without them your card runs once, top to bottom, just like Levels 3 to 10.
3. **Pictures first.** Open the **Animation** tab, press **+**, then pick from the library or draw your own. Give it a short name, such as `boat`.
4. Back in **Code**: `boat.setAnimation("boat");` — the name in quotes matches the Animation tab exactly. A sprite with no animation shows as a grey box.
5. Build in layers on the 400 × 400 canvas: background, shapes, sprites, then text. The **Drawing**, **Sprites**, **Variables** and **Math** drawers hold everything from Levels 1 to 10.

A Whole Card from Levels 1 to 10

A card for a grandad who sails — every line is something you know

```
// LAYER 1 - background and scenery (Levels 3 and 4)
background("lightblue");
fill("navy");
rect(0, 270, 400, 130);             // the sea, bottom third
fill("gold");
ellipse(330, 70, 80, 80);           // the sun

// LAYER 2 - sprites, sized and tilted (Levels 5, 6, 8 and 9)
var boatX = 170;
var boat = createSprite(boatX, 240);
boat.setAnimation("boat");
boat.scale = 0.6;
boat.rotation = -5;                 // leaning into the wind

var gull1 = createSprite(randomNumber(40, 360), randomNumber(30, 150));
gull1.setAnimation("gull");
gull1.scale = 0.3;
var gull2 = createSprite(randomNumber(40, 360), randomNumber(30, 150));
gull2.setAnimation("gull");
gull2.scale = 0.3;
drawSprites();

// LAYER 3 - the message, on top of everything (Level 10)
fill("white");
textSize(32);
text("Happy 70th, Grandad!", 40, 350);
```

The words in quotes after `setAnimation` are the names you gave your pictures in the **Animation tab**. The seagulls land somewhere new on every Run, because of `randomNumber`.

What makes it a great card

It is for one person.	Colours, pictures and message all point at the person in your Criterion A brief.
It fills the square.	Something sits in every part of the 400 × 400 canvas, with the message where the eye lands first.
It is never the same twice.	At least one thing is random, so every Run is a new card.
It reads clearly.	The message is big, and its colour stands out from what is behind it.

My card uses all ten levels

- ☐ A **background** and at least **three shapes**, each with its own `fill` colour.
- ☐ At least one **variable** holding a number I use.
- ☐ At least one thing placed with `randomNumber`.
- ☐ **Three sprites**, each with its `scale` or `rotation` set.
- ☐ A **message**, sized with `textSize`, drawn last so it sits on top.
- ☐ I pressed **Run three times**: something changed each time, and every part was visible.